Johnny Adrián García Betancourt

M.Sc. Researcher in Electrical Engineering | Mechatronics Engineer
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Research Profile

Mechatronics engineer and M.Sc. researcher interested in the geometric and physical structures underlying nonlinear dynamical systems and their mathematical properties. My work draws on differential geometry, geometric mechanics, and port-Hamiltonian systems to investigate foundations for modeling and control. I am equally interested in physics-informed, structure-preserving learning and in understanding the mathematical and dynamical structure of learning itself, with robotics as an application domain.

Education

M.Sc. Student in Electrical Engineering – Control Systems Specialization 2025–Present
King Fahd University of Petroleum & Minerals (KFUPM), Saudi Arabia GPA: 3.938/4.00
Expected Completion: January 2027.
Thesis: *Bridging Physical Safety and Active Tolerance in Robotic Manipulators: A Hierarchical Energy-Aware Architecture.*
Selected Coursework: Geometric Control, Classical Mechanics, Advanced Methods for Theoretical Physics.

Graduate Coursework in Automation and Control (Master's Program) 2024
Escuela Superior Politécnica del Litoral (ESPOL), Ecuador Average: 9.68/10
Selected Coursework: Identification and Control of Dynamical Systems, Machine Learning-Based Control, Industrial Robotics and Computer Vision.

B.Sc. in Mechatronics Engineering (Engineer in Mechatronics) 2017–2022
Escuela Superior Politécnica del Litoral (ESPOL), Ecuador Thesis Grade: 9.20/10
Thesis: *Mechanical and Electrical Design, and Automation of an Industrial Liquid Filling Machine.*

Research Experience

Spectral-Hamiltonian Algebraic Power Evaluation (SHAPE Framework) Research Project | KFUPM
Conceived and formulated a physics-guided fault-diagnosis framework combining port-Hamiltonian power relations with differential flatness to construct residuals without a dedicated dynamic observer. Performed the physical simulations, generated the data, and co-designed and implemented the multi-joint CNN in Python/PyTorch with a collaborator, using temporal, CWT, and spectral representations.

Hierarchical Energy-Aware Architecture for Safe Robotic Interaction M.Sc. Thesis | KFUPM
Developing a fault-tolerant control architecture for redundant manipulators that combines port-Hamiltonian modelling and energy tanks with geometric trajectory replanning. The work investigates energy freezing and regulated tank exchange while adapting motion on the configuration manifold as actuator failures progressively restrict the robot's reachable configurations.

Active Fault-Tolerant Flatness-Based Model Predictive Control Conference Manuscript | Revised for Resubmission
Developed an active fault-tolerant MPC formulation that maps actuator degradation into dynamically consistent constraints in flat-output space. The method adapts feasible trajectory tracking as available control authority decreases, using differential flatness to translate physical actuator limits into algebraic constraints suitable for predictive optimization.

Geometric Modelling & State Estimation of an Underwater Glider Coursework & Research Projects | KFUPM
Led the $\mathrm{SE}(3)$-based multibody modelling of an underactuated floating-base glider with internal mass motion and syringe-based variable-ballast actuators modelled as articulated mechanisms. Used twists and adjoint transformations to model vehicle–mechanism coupling and configuration-dependent inertia; explored gliding and helical motion in MATLAB. In a related Simulink project, implemented Kalman filtering for depth and vertical-velocity estimation.

Industrial Process Automation & Condition Monitoring Graduate Project | ESPOL / Detecuador
Led the design and commissioning of an approximately USD 25,000 liquid-processing project at Detecuador, also developed as graduate academic work at ESPOL. Implemented recipe-based mixing, PLC/HMI automation, tank-level monitoring, and IO-Link communication. Built a separate motor condition-monitoring test bench to study vibration, temperature, and rotational speed; explored MATLAB predictive-maintenance workflows, which were not deployed in the plant.

Design, Modelling & Automation of an Industrial Liquid-Filling Machine B.Sc. Thesis | ESPOL
Designed and modelled an automated filling machine for viscous liquid products, integrating pneumatic actuation, fluid-flow calculations, electrical sizing, PLC/HMI control, and mechanical design. Performed coupled pneumatic/fluid calculations for dosing accuracy and cycle timing, and used CAD/CAE and finite-element analysis to dimension the liquid hopper and structural components.

Research Manuscripts & Working Papers

[1] J. Garcia and N. Salguero, "Active Fault-Tolerant Flatness-Based MPC for Robotic Manipulators under Parametric Mismatch and Actuator Degradation," *Conference Manuscript*, revised for resubmission, 2026.

[2] J. Garcia, N. Salguero, N. Iqbal, B. Brahmi, and R. Rashad, "Energy-Based Fault Detection and Isolation Framework for Flat Port-Hamiltonian Mechanical Systems," *arXiv preprint* arXiv:2610.33679, 2026 (submitted to *IEEE Transactions on Control Systems Technology*).

[3] U. I. Castilla and J. A. Garcia, "Techno-Economic Feasibility of Wind-Powered Hydrogen Production in Saudi Arabia: A Multi-Regional Assessment," *Conference Manuscript*, revised for resubmission, 2026.

Technical Expertise

Mathematical Foundations	Linear and tensor algebra, differential geometry, ODEs/PDEs, calculus of variations, probability and statistics, and mathematical optimization.
Physical Foundations	Classical and geometric mechanics, Lagrangian/Hamiltonian formulations, symplectic and Poisson structures, conservation laws and symmetries, port-Hamiltonian systems, and passivity.
Control & Robotics	Nonlinear and geometric control, stability analysis, differential flatness, optimal control and MPC, robotic dynamics, trajectory optimization, and fault-tolerant control. Screw-theoretic kinematics and the recursive Newton–Euler algorithm (RNEA) for inverse dynamics in Kinova/Franka models.
Machine Learning & AI	Physics-guided machine learning and feature engineering, deep neural networks (1D/2D CNNs), multi-joint feature fusion, model benchmarking, robustness evaluation, and uncertainty quantification.
State Estimation & Signal Processing	Bayesian and invariant state estimation, Lie-group methods, Fourier and time-frequency analysis, digital filtering, image processing, and feature extraction.
Programming & Tools	Python, PyTorch, MATLAB/Simulink, Simscape Multibody, Git, LaTeX, SolidWorks, and ANSYS.
Robotic Platforms	Quanser QArm: vision-based grasping; Kawasaki RS03N at ESPOL: Python-based pick-and-place; Franka Emika: basic hardware motion tests.

Academic and Professional Experience

Teaching Assistant – Electronics and Instrumentation Labs 2025–Present
King Fahd University of Petroleum & Minerals (KFUPM), Saudi Arabia
Led undergraduate laboratory sessions on circuits, signal conditioning and electronic measurements.

Teaching Assistant – Statistics and Probability 2022
Escuela Superior Politécnica del Litoral (ESPOL), Ecuador
Led undergraduate problem-solving sessions; authored midterm and final exam problems and graded assignments.

Maintenance and Automation Coordinator 2021–2024
Detecuador S.A., Ecuador
Led industrial modernization; developed CAD designs, automation, instrumentation and condition-monitoring systems.

Assistant Electromechanical Designer 2023–2024
SEOR S.A., Ecuador
Developed electromechanical CAD designs, control cabinets and prototypes; integrated sensors and programmed PLCs.

Honors and Awards

Graduate Full Scholarship, KFUPM	2025–Present
Best Mechatronics Thesis Award (1st Place), ESPOL	2022
Best University Thesis Award (2nd Place Institution-Wide), ESPOL	2022
Academic Excellence Recognition (Top Score), National University Admission Exam (SENESCYT)	2017

Languages

Spanish – Native **English** – Advanced / Academic working proficiency

Academic References

Dr. Ramy Rashad
Assistant Professor
CIE, KFUPM
M.Sc. Co-advisor
& Geometric Control Instructor

Dr. Brahim Brahmi
Assistant Professor
EE, KFUPM
M.Sc. Thesis Advisor
& Optimal Control Instructor

Dr. Naveed Iqbal
Assistant Professor
EE, KFUPM
Image Processing Instructor
& Research Co-author